SIMATS ENGINEERING

ASSESSMENT TOOL 2: Concept Mapping

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions.* (BL2)

Assessment Tool Weightage: 20%

Dave's Taxonomy: Imitation (Level 1) **Total Marks: 20**

Question Count: 4

Code of Conduct

I Meenakshy V S 192325133 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: _____

Assessment Tasks

Concept Mapping Tasks

Create concept maps for the following tasks. Each map must show key nodes, link labels, and a logical hierarchy.

1. Concept Map 1: Cloud Computing Fundamentals. Include definition, characteristics, benefits, and limitations.
2. Concept Map 2: Evolution of Cloud Computing. Connect hardware evolution, internet software evolution, distributed systems, and service delivery.
3. Concept Map 3: Service Models. Map IaaS, PaaS, SaaS, and XaaS with provider responsibilities and user responsibilities.
4. Concept Map 4: Virtualization and Web Services as Enablers of Cloud. Show how these two concepts support cloud deployment and scalability.

Expected concept nodes:

- Elasticity, scalability, resource pooling, on-demand access, measured service
- Virtual machine, hypervisor, abstraction, API, SOAP, REST, provider, consumer
- Infrastructure layer, platform layer, application layer, shared responsibility

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Concept Mapping Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Concept Identification	25%	All major concepts are accurately identified	Most concepts are identified correctly	Limited concepts are identified	Essential concepts are missing
Relationship Mapping	25%	Links between concepts are accurate and meaningful	Most links are correct	Some links are weak or unclear	Relationships are mostly incorrect
Hierarchy and Organization	20%	Excellent structure with clear hierarchy	Mostly organized with minor issues	Basic structure with weak hierarchy	No logical organization
Technical Relevance	20%	Map strongly reflects cloud course content	Mostly aligned to topic	Partially aligned to topic	Weak alignment to topic
Visual Clarity	10%	Neat, readable, and easy to interpret	Generally readable	Somewhat cluttered	Poorly presented

CO1 ASSESSMENT TOOL - 1

CSAIS20 Cloud Computing : AT1 Concept Mapping

Concept Map 1 : Cloud Computing Fundamentals

Concept Map 2 : Evolution of cloud Computing

Concept Map 3 : Service Models

Concept Map 4 : Virtualization and Web Services as Enable of Cloud.

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Submission date : 09-10-2026

Cloud Computing Fundamentals

CLOUD COMPUPING

A model for delivering Computing resource like Severs, storage, database, software and Serve. over Enfanet on a pay as yougo basis.

Characteristics

- x On demand Self Serve
- x Board network acces
- x Resource Pooling
- x Rapid elasticity
- x Measured Service

Benefits

- x Cost efficency
- x Scalability & flexibility
- x High availability
- x High reliability
- x Faster time to market
- x Reduce IT manage

Limitations

- x Security and Privacy
- x Dependence on Enfant
- x Vendor lock in
- x Limited Control
- x Compliance and data residency issue

Cloud Computing is a Technology that provides Computing resource such as Sever storage applications and Severces over the Enter.

it offer on-demand access, Scalability resource shay rapid elastry and measure service its main benefit. Include cost efficency flexibility high availability and faster deployment. However it also has limitations such as security and privacy risk Enfanet dependency Vendor lock in and limited control over Infactre.

Evolution of Cloud Computing

EVOLUTION OF CLOUD COMPUTING

Hardware Evolution
- Mainframe
- client Server
- Personal Computer
- Virtualization - ready hardware

Internet Software Evolution
- Static website
- Dynamic web app
- web2.0/collaborative application

Distributed Systems
- cluster Computing
- Grid Computing
- Distributed database
- Microservice

Service Delivery
- IaaS
- PaaS
- SaaS
- XaaS

Modern cloud Computing

× Time sharing
→ multiple use share Computer resource
→ Improve resource utilization
→ Reduced Computing cost

× Virtualization
→ create VM
→ improved resource utalization
→ Reduce Computing cost

→ Internet growth, web based Applications
→ enable remote access to resource
→ Cloud Native Technology like
× Container × microservice
× Serverless computing × Devops
→ Cloud feature such as
:- demand access
:- global availability
:- Elastic scalability
:- pay as you go proxy

Key Driver of Cloud Evolution
× Cost Reduction
× Internet Expansion
× Virtualization
× Demand for Scalability
× Automation
× Global Accessibility

Virtualization and Web Service as Enablers of Cloud

Cloud Deployment & Scalability

Virtualization

- Multiple VMs on physical
- Resource isolation
- Efficient resource utilization
- Dynamic provisioning

Web Service

- Standardized API
- Interoperability
- Loose coupling
- Service discovery & integration

Cloud deployment

through abstraction & resource pools

Scalability

through modular, flexible & interconnected

Together they

Support the cloud ability to deploy quickly scale resources on demand and provide flexible

Elasticity

- Scale resource up/down
- Handle changing demand
- Improve efficiency

Resource Pooling

- Shared physical resource
- Better utilization
- Lower cost

Service Models

IaaS
Infrastructure as a Service

- Virtual Machine
- Storage
- Networking
- Computing resource.

Provider responsibility
- physical infrastructure
- Virtualization
- Networking
- Security & availability

User responsibility
- Os, application, data
- Configuration & Management
- Security (in app level)

PaaS
Platform as a Service

- Development platform
- Runtime environment
- Middleware
- Pools & database

Provide responsibility
- Infrastructure
- Os, runtime, middleware
- Security & availability

User responsibility
- Application Code
- Data
- Configuration (applevel)

SaaS
Software as a Service

- Ready to use application
- Web based access
- No Installation

Provider Responsibility
- Application management
- Infrastructure & platforms
- Security & Update

User responsibility
- Use the application
- Manage data & Settings
- Access Control

XaaS
Anything as a Service.

- Any IT Capability
  eg DBaaS, FaaS, BaaS

Provide Responsibility
- Infrastructure / platform
- Service management
- Security & availability

User responsibility
- User the Specific Service
- Manage data/Config
- Follow Service policies.